

Exp 6: Breaking an Affine cipher

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
int main() {
```

```
    char cipher[1000];
```

```
    printf("Enter Ciphertext: ");
```

```
    fgets(cipher, sizeof(cipher), stdin);
```

```
    int a_inv = 9;
```

```
    int b = 15;
```

```
    printf("Recovered Plaintext: ");
```

```
    for (int i = 0; cipher[i] != '\0'; i++) {
```

```
        if (isupper(cipher[i])) {
```

```
            int c = cipher[i] - 'A';
```

```
            int p = (a_inv * (c - b + 26)) % 26;
```

```
            printf("%c", p + 'A');
```

```
        }
```

```
        else if (islower(cipher[i])) {
```

```
            int c = cipher[i] - 'a';
```

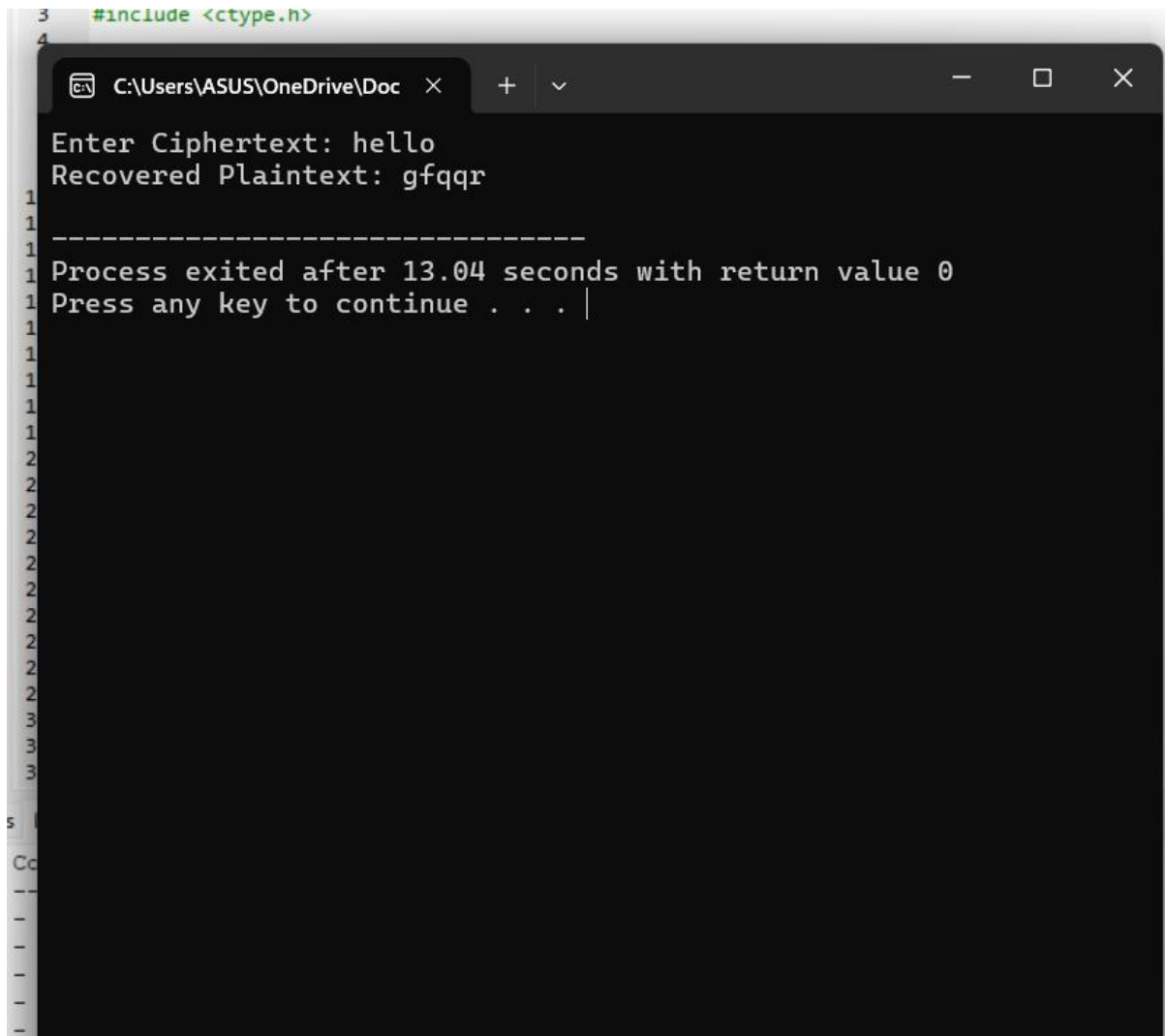
```
            int p = (a_inv * (c - b + 26)) % 26;
```

```
            printf("%c", p + 'a');
```

```
        }
```

```
    else {
```

```
        printf("%c", cipher[i]);  
    }  
}  
  
return 0;  
}
```



```
3  #include <ctype.h>  
4  
C:\Users\ASUS\OneDrive\Doc  ×  +  ▾  -  □  ×  
Enter Ciphertext: hello  
Recovered Plaintext: gfqqr  
-----  
Process exited after 13.04 seconds with return value 0  
Press any key to continue . . . |  
1  
1  
1  
1  
1  
1  
1  
1  
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```